Experiment 14: Grammatical Agreement Checking using CFG in Python

1. Aim

To write a Python program to check for grammatical agreement in sentences based on a context-free grammar's rules.

2. Procedure

1. Import `CFG` and `ChartParser` from the `nltk` module.
2. Define CFG production rules enforcing singular (`NP_SG VP_SG`) and plural (`NP_PL VP_PL`) agreement.
3. Initialize the `ChartParser` with the grammar.
4. Parse test sentences and check whether a valid parse tree is produced.
5. Display the grammatical agreement result for each sentence.

3. Program

```
import nltk
from nltk import CFG

grammar = CFG.fromstring("S -> NP_SG VP_SG | NP_PL VP_PL\nNP_SG -> 'dog'\nNP_PL -> 'dogs'\nVP_SG -> 'barks'\nVP_PL -> 'bark'")
parser = nltk.ChartParser(grammar)

for s in ["dog barks", "dogs bark", "dog bark"]:
    valid = len(list(parser.parse(s.split()))) > 0
    print(f"'{s}' ->", "Agreement Valid" if valid else "Agreement Invalid")
```

4. Output

```
'dog barks' -> Agreement Valid
'dogs bark' -> Agreement Valid
'dog bark' -> Agreement Invalid
```

5. Result

Thus, the Python program was successfully executed to check for agreement in sentences based on a context-free grammar's rules.